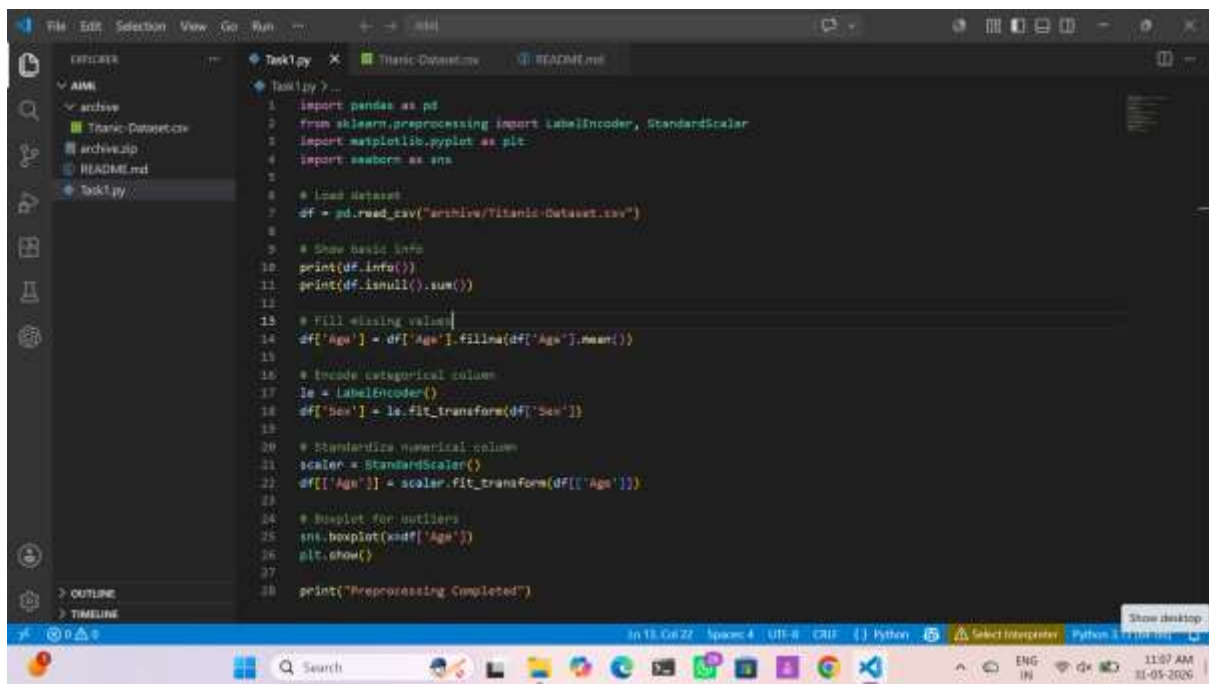


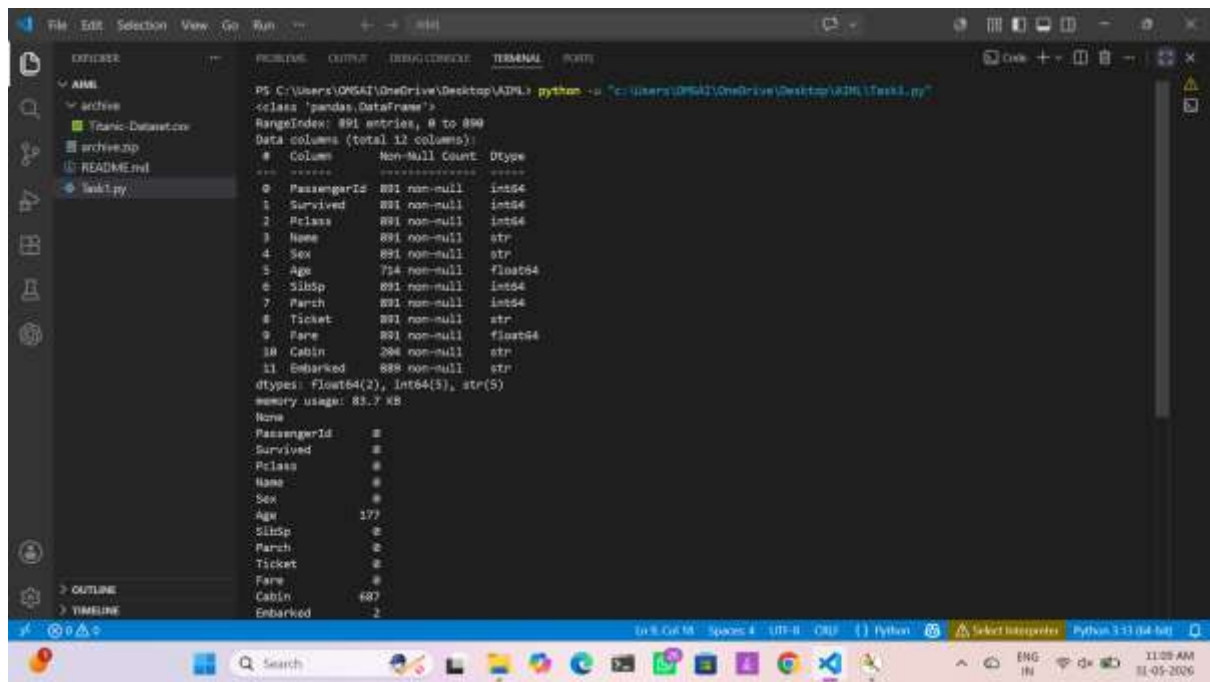
CODE :-



The image shows a Jupyter Notebook titled 'Task1.py' open in a VS Code editor. The notebook contains Python code for loading, cleaning, and preprocessing the Titanic dataset. The code includes imports for pandas, sklearn, matplotlib, and seaborn. It loads the dataset from a CSV file, displays basic information, fills missing values in the 'Age' column, encodes the 'Sex' column, and standardizes the 'Age' column. Finally, it creates a boxplot for outliers and prints a completion message.

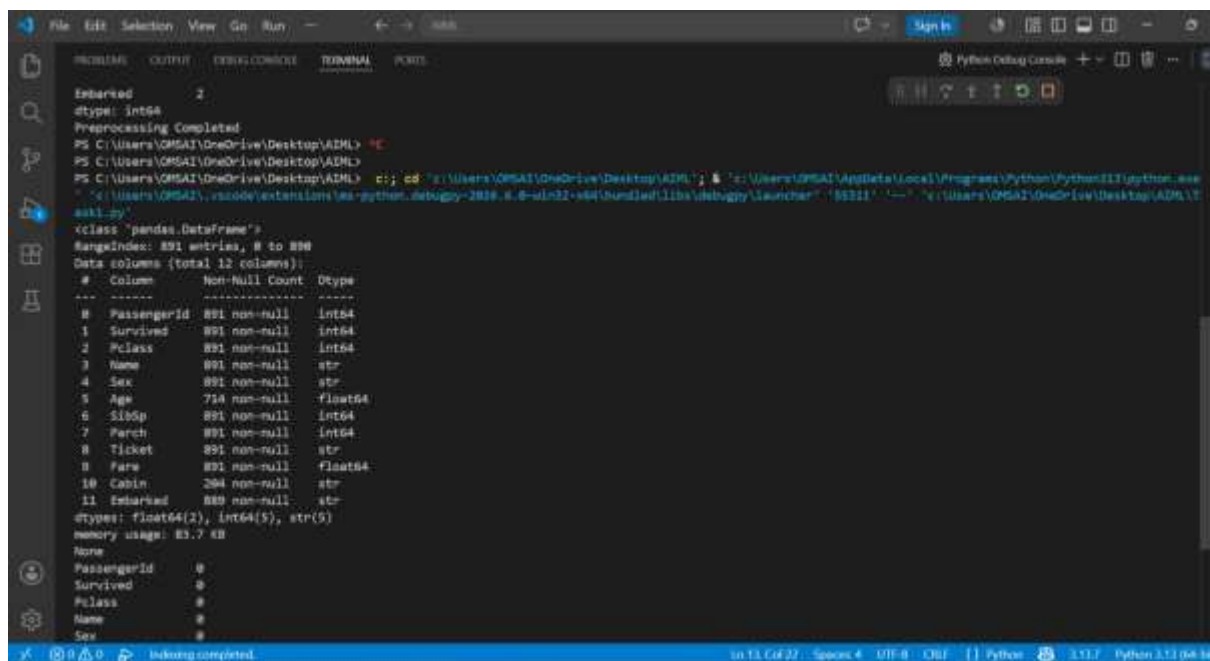
```
1 import pandas as pd
2 from sklearn.preprocessing import LabelEncoder, StandardScaler
3 import matplotlib.pyplot as plt
4 import seaborn as sns
5
6 # Load dataset
7 df = pd.read_csv("archive/Titanic-Dataset.csv")
8
9 # Show basic info
10 print(df.info())
11 print(df.isnull().sum())
12
13 # Fill missing values
14 df['Age'] = df['Age'].fillna(df['Age'].mean())
15
16 # Encode categorical column
17 le = LabelEncoder()
18 df['Sex'] = le.fit_transform(df['Sex'])
19
20 # Standardize numerical column
21 scaler = StandardScaler()
22 df[['Age']] = scaler.fit_transform(df[['Age']])
23
24 # Boxplot for outliers
25 sns.boxplot(x=df['Age'])
26 plt.show()
27
28 print("Preprocessing Completed")
```

OUTPUT:-



```
PS C:\Users\OMSAI\OneDrive\Desktop\ADML> python -u "c:\Users\OMSAI\OneDrive\Desktop\ADML\Task1.py"
<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column             Non-Null Count  Dtype  
---  --
 0   PassengerId        891 non-null    int64  
 1   Survived           891 non-null    int64  
 2   Pclass            891 non-null    int64  
 3   Name              891 non-null    str     
 4   Sex               891 non-null    str     
 5   Age              714 non-null    float64 
 6   SibSp            891 non-null    int64  
 7   Parch            891 non-null    int64  
 8   Ticket           891 non-null    str     
 9   Fare             891 non-null    float64 
10   Cabin            204 non-null    str     
11   Embarked          889 non-null    str     
dtypes: float64(2), int64(5), str(5)
memory usage: 83.7 KB

None
PassengerId    0
Survived       0
Pclass         0
Name           0
Sex            0
Age           177
SibSp          0
Parch          0
Ticket         0
Fare           0
Cabin         680
Embarked       2
```



```
Embarked       2
dtype: int64
Preprocessing Completed
PS C:\Users\OMSAI\OneDrive\Desktop\ADML> cd "c:\Users\OMSAI\OneDrive\Desktop\ADML" & "c:\Users\OMSAI\AppData\Local\Programs\Python\Python311\python.exe" -u "c:\Users\OMSAI\OneDrive\Desktop\ADML\Task1.py"
<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column             Non-Null Count  Dtype  
---  --
 0   PassengerId        891 non-null    int64  
 1   Survived           891 non-null    int64  
 2   Pclass            891 non-null    int64  
 3   Name              891 non-null    str     
 4   Sex               891 non-null    str     
 5   Age              714 non-null    float64 
 6   SibSp            891 non-null    int64  
 7   Parch            891 non-null    int64  
 8   Ticket           891 non-null    str     
 9   Fare             891 non-null    float64 
10   Cabin            204 non-null    str     
11   Embarked          889 non-null    str     
dtypes: float64(2), int64(5), str(5)
memory usage: 83.7 KB

None
PassengerId    0
Survived       0
Pclass         0
Name           0
Sex            0
```

GRAPH:-

